
Target-Speaker Speech-to-Text in Overlapping Speech

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Abstract

Modern speech-to-text systems achieve low word error rates on clean, single-speaker audio but degrade sharply when speakers overlap. In this technical report, we present a target-speaker speech-to-text system that, given a short enrollment clip of the desired speaker and a multi-speaker audio mixture, transcribes only the target speaker. Our pipeline combines a frozen pretrained speaker verification model (ECAPA-TDNN) to extract a speaker embedding, a trained speaker-conditioned masking network that isolates the target speaker’s spectrogram, and a frozen pretrained ASR model (Whisper) for transcription. We evaluate on synthetic two-speaker mixtures from LibriSpeech at varying signal-to-interferer ratios and overlap ratios, measuring word error rate. We show that overlapping speech degrades ASR performance across languages, and present our extraction-based approach to mitigate this degradation, achieving up to 32 percentage point absolute and 48% relative reductions in WER.

Attestation of Teamwork

Jibran Iqbal Shah: Designed and trained the speaker-conditioned extraction network, including architecture design, loss function development (power-law compression, hybrid oracle mask loss), hyperparameter tuning, and all related code and writeup.

Yamoah Frimpong Attafuah: Fine-tuned Whisper-small on the WAXAL Akan dataset for Twi speech recognition, set up the WER evaluation scripts and notebooks, and benchmarked both the baseline Whisper and fine-tuned Whisper models across varying overlap ratios and SIR levels for English and Twi. Wrote the code and writeup related to these contributions.

1 Introduction

Modern speech-to-text systems perform well on clean, single-speaker audio but degrade sharply when speakers overlap [Watanabe et al., 2020, Kanda et al., 2020]. This is a common occurrence in meetings, court proceedings, and media such as films or talk shows. Rather than attempting full diarization and separation, target-speaker transcription offers a more focused alternative: given a short enrollment clip, transcribe only the speaker of interest.

Additionally, the rise of full-duplex speech-to-speech models is increasing the demand for labelled transcriptions of natural conversations with interruptions [Défossez et al., 2024, Roy et al., 2026]. A

reliable target-speaker system could automatically generate such labels by isolating each participant and producing timed transcripts, even through overlapping speech.

The objective of this project is to develop a target-speaker speech-to-text system that takes (i) a short enrollment utterance of the target speaker and (ii) a multi-speaker mixture, and outputs only the target speaker’s transcript. We test across English and Twi to examine whether the degradation pattern is language-agnostic.

2 Preliminaries and Problem Formulation

Let $\mathbf{x}_{\text{mix}} \in \mathbb{R}^T$ be a single-channel audio mixture containing overlapping speech from $N \geq 2$ speakers. We are also given a short enrollment utterance $\mathbf{x}_{\text{enroll}} \in \mathbb{R}^{T_e}$ of the target speaker s^* . The goal is to produce a transcript $\hat{y}$ that matches the ground-truth transcript y^* of speaker s^* within the mixture, as measured by word error rate $\text{WER} = (S + D + I)/N_{\text{ref}}$, where S, D, I are the number of substitutions, deletions, and insertions and N_{ref} is the number of words in the reference transcript. Note that WER can exceed 100% when the model produces more insertions than there are reference words.

The Short-Time Fourier Transform (STFT) converts a waveform into a two-dimensional complex-valued matrix $\mathbf{X} \in \mathbb{C}^{F \times T}$, where each entry encodes the amplitude and phase of frequency bin f during time frame t . The magnitude spectrogram $|\mathbf{X}| \in \mathbb{R}^{F \times T}$ retains full linear frequency resolution, which is needed to invert back to a waveform after masking via the inverse STFT (iSTFT). We use standard parameters for 16 kHz speech: 25 ms window (400 samples), 10 ms hop (160 samples), FFT size $N_{\text{fft}} = 512$, giving $F = 257$ frequency bins linearly spaced from 0–8000 Hz.

3 Design

Our pipeline consists of three stages, illustrated in Figure 1:

1. Speaker embedding extraction: A frozen pretrained ECAPA-TDNN [Desplanques et al., 2020] encodes the enrollment utterance $\mathbf{x}_{\text{enroll}}$ into a speaker embedding $\mathbf{e} \in \mathbb{R}^{192}$.
2. Speaker-conditioned extraction: A trained masking network receives the mixture magnitude spectrogram $|\mathbf{X}_{\text{mix}}|$ and the speaker embedding $\mathbf{e}$, and predicts a soft mask $\mathbf{M} \in [0, 1]^{F \times T}$ that isolates the target speaker. The estimated target spectrogram is $|\hat{\mathbf{X}}_{\text{target}}| = \mathbf{M} \odot |\mathbf{X}_{\text{mix}}|$, and the waveform is reconstructed via iSTFT using the original mixture phase.
3. ASR decoding: A frozen pretrained Whisper model [Radford et al., 2023] transcribes the extracted waveform.

Only the extraction network (stage 2) is trained; the speaker encoder and ASR model remain frozen throughout.

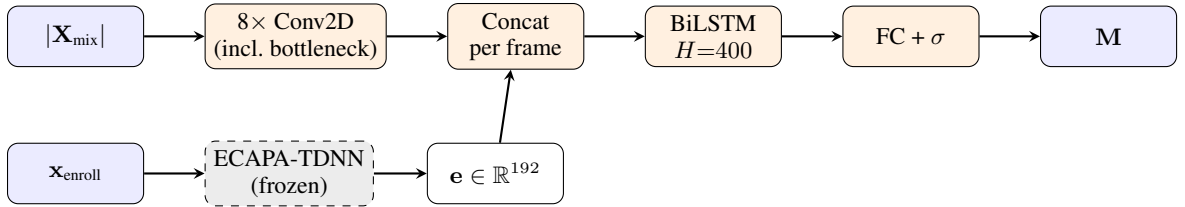


Figure 1: Extraction network architecture. The magnitude spectrogram is encoded by eight 2D convolutional layers, conditioned on the speaker embedding via concatenation, and processed by a BiLSTM to predict a soft mask $\mathbf{M} \in [0, 1]^{F \times T}$. Dashed box denotes a frozen pretrained model.

4 Methodology

4.1 Speaker embedding and extraction network architecture

We use a pretrained ECAPA-TDNN [Desplanques et al., 2020] (speechbrain/spkrec-ecapa-voxceleb) to extract a 192-dimensional speaker embedding from the enrollment utterance. The model achieves 0.80% equal error rate on VoxCeleb1 and is kept frozen throughout.

The extraction network follows the VoiceFilter design [Wang et al., 2019] and has 9.53M trainable parameters. The magnitude spectrogram is treated as a single-channel 2D image and processed by eight Conv2D layers with layer normalization and ReLU. The first two layers use separable kernels (frequency-only and time-only), layers 3–7 use joint (5, 5) kernels with time-only dilation (factors 1, 2, 4, 8, 16) giving a temporal receptive field of 31 frames (≈ 310 ms), and the final layer is a 1×1 bottleneck reducing from 64 to 8 channels.

The CNN output is flattened per time frame and concatenated with the speaker embedding $\mathbf{e} \in \mathbb{R}^{192}$. A single-layer BiLSTM ($H=400$ per direction) processes the sequence, and two FC layers with ReLU and sigmoid produce the mask $\mathbf{M}_t \in [0, 1]^F$ at each frame. We use a BiLSTM rather than a unidirectional LSTM because overlap regions are locally ambiguous and both past and future context help disambiguate which spectral energy belongs to the target speaker.

4.2 Training objective and data pipeline

We train with a power-law compressed reconstruction loss following Wilson et al. [Wilson et al., 2018], who show that element-wise power-law compression of spectrograms before computing MSE improves separation quality. We use an exponent of $p = 0.3$, which we confirmed through a short ablation over $p \in \{0.1, 0.2, 0.3, 0.5, 1.0\}$:

$$\mathcal{L}_{\text{recon}} = \frac{1}{FT} \left\| (\mathbf{M} \odot |\mathbf{X}_{\text{mix}}| + \epsilon)^p - (|\mathbf{X}_{\text{target}}| + \epsilon)^p \right\|_F^2$$

where $\epsilon = 10^{-8}$. The same compression is applied to the CNN input to normalize its dynamic range; the mask is still applied to raw magnitudes for reconstruction.

We also experimented with a hybrid loss that adds direct supervision on the oracle mask. Since the mask should satisfy $\mathbf{M} \odot |\mathbf{X}_{\text{mix}}| = |\mathbf{X}_{\text{target}}|$, dividing both sides by $|\mathbf{X}_{\text{mix}}|$ gives the ideal mask $\mathbf{M}_{f,t}^* = \text{clamp}(|\mathbf{X}_{\text{target}}|_{f,t} / (|\mathbf{X}_{\text{mix}}|_{f,t} + \epsilon), 0, 1)$, clamped to $[0, 1]$ since the sigmoid output cannot exceed this range:

$$\mathcal{L} = \alpha \cdot \|\mathbf{M} - \mathbf{M}^*\|_F^2 + (1 - \alpha) \cdot \mathcal{L}_{\text{recon}}$$

This hybrid approach is inspired by Wilson et al. [Wilson et al., 2018], who explore tradeoffs between mask-domain and signal-domain losses. We found $\alpha = 0.1$ to best balance optimizing the oracle mask loss and the reconstruction loss, through a sweep over $\alpha \in \{0, 0.05, 0.1, 0.2, 0.3, 1.0\}$.

We use LibriSpeech [Panayotov et al., 2015] (dev-clean + test-clean, 16 kHz mono) for English and WAXAL Akan [Diack et al., 2026] for Twi. We generate 12,500 synthetic two-speaker mixtures from LibriSpeech: $\mathbf{x}_{\text{mix}} = \mathbf{x}_{\text{target}} + \alpha \mathbf{x}_{\text{interferer}}$, varying SIR $\in \{0, 5\}$ dB and overlap ratio $\in \{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$. Each mixture preserves the aligned clean target waveform for computing oracle masks during training. The recipes are split into 10,000 train, 1,250 validation, and 1,250 test samples, with speaker embeddings precomputed for all 2,588 unique enrollment utterances.

5 Numerical Experiments

5.1 Whisper fine-tuning for Twi

Whisper does not natively support Twi (Akan), producing near-100% WER and CER out of the box. We fine-tuned Whisper-small (244M params) on the WAXAL Akan dataset [Diack et al., 2026], reducing clean-speech WER from $\sim 100\%$ to 33.3% and CER from $\sim 100\%$ to 12.1%. This fine-tuned model serves as the Twi baseline for our overlap experiments.

5.2 Baseline: overlap degrades ASR across languages

We evaluated Whisper-tiny (39M params) on English and the fine-tuned Whisper-small on Twi across all SIR levels and overlap ratios.

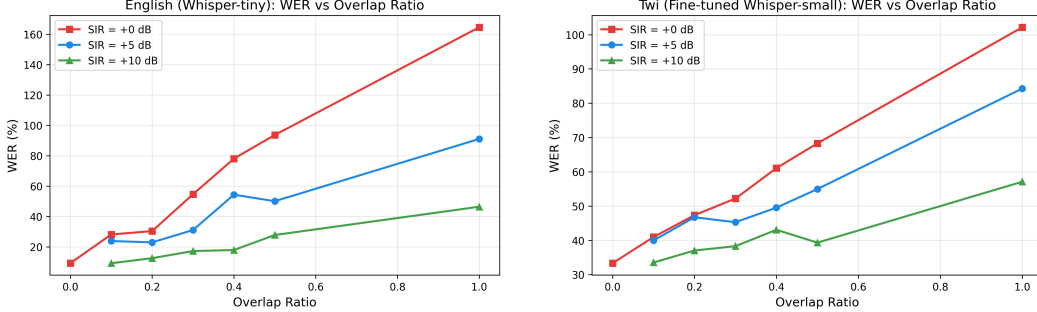


Figure 2: WER vs. overlap ratio for English (left, Whisper-tiny, clean baseline 8.1%) and Twi (right, fine-tuned Whisper-small, clean baseline 34.7%) at three SIR levels. Both languages show the same degradation pattern: WER increases with overlap ratio and decreases with SIR. At 0 dB SIR with full overlap, English WER exceeds 100%. This confirms the problem is language-agnostic.

5.3 Training and WER evaluation

We trained with batch size 16 (maximum that fits in GPU memory with 14-second random crops), initial learning rate 10^{-3} , Adam optimizer, hybrid loss $\alpha = 0.1$, power-law compression $p = 0.3$, and early stopping with patience 7. The learning rate is halved by ReduceLROnPlateau whenever validation loss does not improve for 3 consecutive epochs.

We evaluated seven trained configurations on the 1,250-sample test set, including a hybrid loss model ($\alpha = 0.1$) and a power-law reconstruction loss model ($\alpha = 0$); curriculum-based variants are described in the next section. All models were trained until validation loss plateaued.

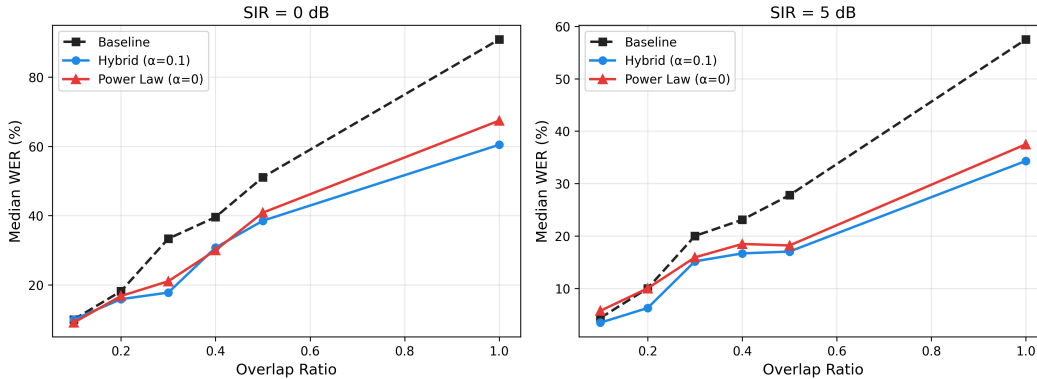


Figure 3: Median WER vs. overlap ratio for baseline (no extraction), hybrid loss ($\alpha = 0.1$), and power-law reconstruction loss only ($\alpha = 0$) at 0 dB SIR (left) and 5 dB SIR (right). Both extraction models reduce WER at overlap ratios ≥ 0.3 , with the largest gains at full overlap.

Both extraction models significantly reduce median WER compared to the baseline at overlap ratios ≥ 0.3 . At full overlap (1.0), the hybrid model reduces WER from 90.9% to 60.5% at 0 dB and from 57.5% to 34.3% at 5 dB. The power-law model achieves smaller but still notable gains (67.4% and 37.5% respectively). The hybrid model consistently outperforms power-law across most overlap conditions, with the gap widening at mid-range overlaps (0.3–0.5) where the oracle mask supervision

prevents the pass-through shortcut. At low overlap (≤ 0.2), extraction provides marginal or no benefit, as the interference is brief enough for Whisper to handle without assistance.

5.4 Curriculum learning

Curriculum learning trains a model on progressively harder examples rather than sampling the full difficulty range uniformly. The idea is that easy examples teach the model the basic task first, and harder examples are introduced once that foundation is in place.

Following the VoiceFilter paper [Wang et al., 2019], we first tried training only on hard examples by restricting the training set to overlap $\in \{0.5, 1.0\}$. The results were poor: the Hard-dist. column of Table 1 shows near-zero improvement and slight degradation at full overlap. Low-overlap examples provide clean, unambiguous oracle masks that the model needs to learn basic masking behavior; without them, training never gets off the ground.

We therefore used a three-phase curriculum. In Phase 1 (Easy), the model is trained from scratch on overlap ≤ 0.3 until val loss plateaued, learning to separate well-separated mixtures. In Phase 2 (Medium), training resumes from that checkpoint with the harder ratios 0.4 and 0.5 added, upsampled at weight $2\times$, and continues until val loss plateaued again. In Phase 3 (Hard), training resumes from the Medium checkpoint on the full overlap range $\{0.1, 0.2, 0.3, 0.4, 0.5, 1.0\}$ with upsampling weights $[1, 1, 1, 2, 2, 3]$, putting more gradient on the hardest conditions. All phases use hybrid loss ($\alpha = 0.1$).

Table 1 compares median WER across all seven trained configurations. Curriculum Hard achieves the best result on nearly every row, with the largest gains at mid-range overlaps (0.2–0.5) where the progressive upsampling had the most effect.

Table 1: Median WER (%) for all models across SIR and overlap conditions. **Bold** indicates the best extraction model per row. *Curriculum Easy* was trained on overlap ≤ 0.3 only; *Curriculum Medium* extends to overlap ≤ 0.5 via warm-start fine-tuning; *Curriculum Hard* resumes from Medium and trains across all overlaps with weights $[1, 1, 1, 2, 2, 3]$.

SIR (dB)	Overlap	Baseline	Hard-dist. (PL, $\{0.5, 1.0\}$)	Power Law ($\alpha = 0$)	Hybrid ($\alpha = 0.1$)	Curriculum Easy	Curriculum Medium	Curriculum Hard
0	0.1	10.0	9.5	9.1	10.0	9.1	9.1	9.1
0	0.2	18.2	18.2	16.8	15.9	18.3	16.2	11.0
0	0.3	33.3	28.1	21.1	17.8	22.2	20.0	17.2
0	0.4	39.6	37.5	30.0	30.8	37.5	29.2	26.9
0	0.5	51.1	48.0	40.9	38.5	43.7	36.5	34.0
0	1.0	90.9	93.8	67.4	60.5	84.6	61.5	58.8
5	0.1	4.3	4.8	5.7	3.4	4.8	3.6	3.4
5	0.2	10.0	10.0	10.0	6.2	9.5	8.7	6.7
5	0.3	20.0	20.0	15.9	15.2	17.4	15.4	14.8
5	0.4	23.1	22.2	18.5	16.7	20.0	17.9	13.5
5	0.5	27.8	25.0	18.2	17.0	23.8	16.7	16.7
5	1.0	57.5	50.0	37.5	34.3	48.9	36.0	34.3

Models trained without oracle mask supervision predict degenerate masks: Power Law ($\alpha = 0$) converges to mask ≈ 1.0 (mean 0.91), passing the mixture through unchanged, while Hard-dist. settles near 0.5 (mean 0.49), uniformly attenuating all frequencies. Models trained with oracle mask supervision (Hybrid, Curriculum) produce bimodal masks (mean ≈ 0.36 , median = 0.0), with $\sim 59\%$ of values near zero (suppressing interferer bins) and $\sim 30\%$ near one (preserving target bins), reflecting genuine speaker separation rather than a shortcut.

6 Discussion

The extraction network reduces WER most where baseline Whisper struggles most: at high overlap ratios and low SIR. At 0 dB SIR with full overlap, median WER drops from 90.9% to 58.8% with the best model, a 32 percentage point reduction. At 5 dB SIR with full overlap the reduction is from 57.5% to 34.3%. The greatest relative gain is at 0.3 overlap (0 dB SIR), where WER falls by 48.3% relative to the unprocessed mixture. However, the extracted WER at full overlap remains

far from the clean-speech baseline of 8.1%. We also verified that running extraction on clean speech (no interferer, overlap ≈ 0) causes no noticeable WER degradation across all trained models, confirming the extractor does not suppress the target signal when no interference is present. Mask distribution analysis confirms that the hybrid loss is beneficial: power-law-only training collapses to a near-uniform mask, while the hybrid loss produces bimodal masks with genuine suppression of the interfering speaker. Curriculum learning further improves over uniform sampling from the full training set. Our reductions are comparable to VoiceFilter, which also reduces WER by roughly 58% in the best case, though on a larger training set. VoiceFilter’s training distribution did not use curriculum ordering, and our results suggest it may have achieved even better gains had it adopted a similar progressive approach.

We demonstrated that overlapping speech degrades ASR in both English and Twi, confirming the problem is language-agnostic. The extraction network operates on spectrograms and speaker embeddings rather than language-specific features, so it should in principle generalize across languages. We did not have time to evaluate the extraction network on Twi, but this is a natural next step: training on English and testing on Twi would reveal whether the learned masking strategy transfers cross-lingually without additional fine-tuning.

Qualitative listening examples in the demo notebook show that in some cases the interferer is substantially dampened in the extracted audio. This dampening already happens implicitly: the oracle mask pushes the predicted mask toward zero in bins where the interferer dominates, so interferer suppression is a side effect of mask supervision rather than an explicit objective. A more direct objective is to maximise the output Signal-to-Interference Ratio (SIR) explicitly, for example:

$$\mathcal{L}_{\text{SIR}} = \log\|\mathbf{M} \odot |\mathbf{X}_{\text{interf}}|\|^2 - \log\|\mathbf{M} \odot |\mathbf{X}_{\text{target}}|\|^2$$

where $|\mathbf{X}_{\text{interf}}|$ and $|\mathbf{X}_{\text{target}}|$ are the interferer and target magnitude spectrograms (both available at training time since the mixtures are synthetic). Unlike a simple energy penalty on the interferer (i.e., only the first term), this ratio form has a positive signal (pass target energy) and a negative signal (block interferer energy), so it cannot be trivially minimised by zeroing the mask entirely. Along another loss direction, Wu et al. [Wu et al., 2025] propose adding a speaker consistency term $\mathcal{L}_{\text{sc}} = 1 - \cos(\mathbf{e}_{\text{enroll}}, \hat{\mathbf{e}}_{\text{extracted}})$ based on the cosine similarity between the speaker embedding of the extracted waveform and the enrollment audio. This would force the model to make fuller use of the enrollment embedding, since the extracted output would need to be acoustically close to the enrolled speaker and not just minimise reconstruction error. On the architecture side, our current design conditions the masking network on a single global 192-dimensional speaker embedding concatenated per frame. Recent work such as USEF-TSE [Zeng and Li, 2024] replaces this with multi-head cross-attention over frame-level enrollment features, allowing the model to attend to fine-grained speaker characteristics at each time step rather than compressing them into a fixed vector. This could improve separation quality, particularly in challenging full-overlap conditions.

7 Conclusions

We built a target-speaker speech-to-text system combining a frozen ECAPA-TDNN speaker encoder, a trained masking network, and a frozen Whisper ASR model. The extraction network reduces median WER by up to 32 percentage points at full overlap (0 dB SIR) compared to running Whisper directly on the mixture and reduces the relative error by nearly 48% at lower overlaps. Speaker overlap degrades ASR in both English and Twi, and the extraction approach partially recovers the target signal in English. Training with a hybrid loss ($\alpha = 0.1$) combining power-law reconstruction loss with oracle mask supervision works best. A three-phase curriculum that progressively introduces harder overlap ratios further improves performance, particularly at mid-range overlaps (0.2–0.5), and strictly outperforms all non-curriculum baselines.

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Use of AI

We used Claude (Anthropic) to assist with implementation, debugging, generating plots, and documenting the codebase, as well as to brainstorm ideas for improving model performance. All architecture decisions, experimental design, and analysis were done by the authors. AI-generated code was reviewed and tested before use.

Consent for Information Sharing

- All group members consent to allow the project materials (report, slides, and presentation recording) to be shared on the course page for future students.

Consent for Use of Project Information in Statistical Analysis

- All group members consent to allow anonymized information about the project (e.g., topic, methods, outcomes, grades) to be used by the instructor for statistical analysis and course improvement.

Group Identification

- Group number: 28
- Names of group members: Jibran Iqbal Shah, Yamoah Frimpong Attafuah
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- Date: April 10, 2026